

Computer Networks Basics

Chapter 1: Introduction to Computer Networks

Computer networks connect multiple computers for resource sharing.

Network Types: LAN, MAN, WAN

OSI Model: 7 layers - Physical, Data Link, Network, Transport, Session, Presentation, Application

TCP/IP Protocol Stack: Application, Transport, Internet, Link

Chapter 2: Physical Layer

Transmission media: Twisted pair, Fiber optic, Wireless

Chapter 3: Data Link Layer

Ethernet is the most common LAN technology.

Frame structure and error detection.

Chapter 4: Network Layer

IP protocol provides connectionless, unreliable delivery.

Routing algorithms: Distance vector (RIP), Link state (OSPF)

Chapter 5: Transport Layer

TCP: Reliable, connection-oriented protocol

UDP: Unreliable, connectionless protocol

References: Kurose & Ross, Computer Networking, 8th ed.